SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Project Initialization and Planning Phase

Date	04 July 2026
Team ID	
Project Title	HDI Prediction System – A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	Develop a machine learning-based system to predict the Human Development Index (HDI) of countries using key socio-economic indicators such as life expectancy, education, and income. The system aims to support data-driven policy and development planning.
Scope	The project includes data collection, preprocessing, machine learning model development, HDI prediction, data visualization, and report generation through an interactive web application.
Problem Statement	
Description	Current methods of analyzing HDI rely on historical reports and manual analysis, making it difficult to predict future development trends and identify areas requiring policy intervention.

Impact	An accurate HDI prediction system will help governments, researchers, NGOs, and policymakers make informed decisions, allocate resources effectively, and monitor development progress.
Proposed Solution	
Approach	Use machine learning algorithms to analyze historical socio-economic data and predict HDI values. The application will provide visualizations, prediction reports, and insights to support decision-making.
Key Features	- HDI prediction using machine learning. - Interactive dashboard with charts and graphs. - Country-wise comparison and trend analysis. - Data preprocessing and feature engineering.

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	CPU/GPU	Intel Core i5/Ryzen 5 or above
Memory	RAM specifications	8 GB RAM
Storage	Disk space for data, models, and logs	256 GB SSD or higher
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	Scikit-learn, Pandas, NumPy, Matplotlib, Plotly

Development Environment	IDE	VS Code or PyCharm
Data		
Data	Source, size, format	UNDP/World Bank HDI dataset (CSV format)